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# Optimization of Enhanced Surfaces for High Flux Chip Cooling by Pool Boiling

*A high flux electronic chip was numerically and experimentally simulated to investigate pool boiling capabilities of enhanced metallic surface attachments built upon a  $12.7 \times 12.7 \text{ mm}^2$  base area. It is shown how experimental nucleate boiling data for a flat chip and for chips with low-profile microstructures can be used as input boundary conditions in the numerical prediction of boiling performances of high flux, smooth and microstructured extended cylindrical surfaces. A technique for extending the applicability of the numerical results to cylindrical fin arrays is demonstrated with the aid of experimental data obtained for these surfaces. Surface enhancement resulted in chip planform heat fluxes of 105.4 and 159.3 W/cm<sup>2</sup>, for saturated and 35°C subcooled FC-72, respectively.*

## 1 Introduction

The continuing trend of increasing computer power has driven the number of logic devices etched onto an integrated circuit up at a rapid pace. The quest toward developing lower power logic devices has been overwhelmed by concurrent increases in switching speed and number of devices per unit area. Consequently, more powerful cooling technologies must be developed to help sustain the trends of increased miniaturization and computational power.

Direct liquid immersion cooling has great potential as a successor to air cooling and indirect liquid cooling technologies. Currently, several immersion-cooled computers are in operation. The Cray-2, for example, relies on a slow moving flow of a dielectric fluorocarbon liquid (FC-77) to dissipate heat by forced convection and localized boiling (Danielson et al., 1986). Because of its dielectric properties and chemical inertness, the FC family of fluorocarbons (manufactured by 3M) appears to be the obvious choice for direct immersion cooling of electronic hardware.

While direct immersion has been studied in the context of single-phase forced convection (Tuckerman and Pease, 1981; Samant and Simon, 1986; Ramadhyani and Incropera, 1987; Maddox and Mudawar, 1989; Mudawar and Maddox, 1989), falling films (Mudawar et al., 1987; Grimley et al., 1988), and flow boiling (Samant and Simon, 1986; Maddox and Mudawar, 1989; Mudawar and Maddox, 1989; McGillis et al., 1991), pool boiling remains a very attractive cooling option due to the unique attribute of passive fluid circulation. Having no mechanical pumps, pool boiling hardware are less complex, easier to seal and free of pump-induced fluid pulsation.

Compared to forced convection, pool boiling offers few options for enhancement of heat transfer. Liquid velocity in a flow channel enhances both single and two-phase convection and postpones critical heat flux (CHF) by sweeping away bubbles before they form an insulating vapor blanket. In pool boiling, however, the relatively weak liquid flow is induced only by natural convection prior to the onset of boiling and, thereafter, by buoyancy and bubble motion. Unfortunately, induced flow cannot be easily controlled, therefore, attempts at enhancing pool boiling performance are limited to variations of (1) fluid conditions, type of coolant, saturation pressure, and subcooling (Mudawar and Anderson, 1990); or (2) surface structure, either on a microscopic scale (Messina and Park, 1981; Anderson and Mudawar, 1989) or by a large increase of surface area (Marto and Lepere, 1982; Nakayama et al., 1984; Anderson and Mudawar, 1990).

The planar nature of the electronic chip lends itself to the attachment of extended surfaces to increase surface area and, therefore, the transfer rate from the chip. Several boiling fin design methods have been proposed in the past (Haley and Westwater, 1966; Cumo et al., 1968; Cash et al., 1971), but the specific requirements for electronic cooling were first addressed by Nakayama et al. (1984). Using a cylindrical fin constructed of stacked porous plates, they were able to dissipate over 110 W/cm<sup>2</sup> (based on planform area) in FC-72.

The present study is intended as a general guide to the packaging engineer in designing boiling fins for electronic cooling. This is achieved through adaptation of experimental nucleate boiling data measured for a flat chip and for low-profile microstructured chips in predicting boiling performance of smooth and microstructured extended surfaces, respectively. It is also the objective of this paper to examine means of combining several levels of surface enhancement to boost CHF above the chip fluxes predicted for future very large scale integration (VLSI) chips.

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